Teacher Module - Use Case Narratives

Classroom Management System, Teacher Module (SRS v1.0)

1. Purpose and scope

This document gives a narrative for each use case of the Teacher Module. Each narrative is derived from the teacher user stories US-T1 to US-T14 and the functional requirements FR-01 to FR-30 of the SRS. Log in (UC-T01) is shared by all other teacher use cases.

2. Use case index

ID	Use case	Source	Primary actor
UC-T01	Log in & view dashboard	FR-01, FR-02; US-T1	Teacher
UC-T02	Select course & section	FR-03, FR-04; US-T2	Teacher
UC-T03	Take attendance	FR-05, FR-06, FR-07; US-T3	Teacher
UC-T04	View & update attendance records	FR-08, FR-09; US-T4	Teacher
UC-T05	Post announcement	FR-10, FR-11; US-T5	Teacher
UC-T06	Notify students	FR-12, FR-13; US-T6	System
UC-T07	Create & publish assignment	FR-14, FR-15; US-T7	Teacher
UC-T08	Review student submissions	FR-16, FR-17, FR-18; US-T8	Teacher
UC-T09	Record quiz & exam marks	FR-19, FR-20; US-T9	Teacher
UC-T10	Edit submitted results	FR-21, FR-22; US-T10	Teacher

UC-T11	Submit final grades	FR-23, FR-24; US-T11	Teacher
UC-T12	Provide feedback to students	FR-25, FR-26; US-T12	Teacher
UC-T13	Report student issue	FR-27, FR-28; US-T13	Teacher
UC-T14	View schedule & pending tasks	FR-29, FR-30; US-T14	Teacher

3. Use case narratives

UC-T01: Log in & view dashboard

Use case ID	UC-T01
Use case name	Log in & view dashboard
Source requirements	FR-01, FR-02; US-T1
Primary actor	Teacher
Secondary actors	Authentication service
Description	The teacher authenticates with email and password and reaches a dashboard that summarizes assigned courses, today's classes, pending tasks and recent notifications.
Preconditions	The teacher has a registered account (SRS assumption). The teacher is not logged in.
Trigger	The teacher opens the system login page.
Main success scenario	1. Teacher enters email address and password. 2. System checks the credentials against the stored password hash. 3. System creates an authenticated session. 4. System displays the dashboard with assigned courses, today's classes, pending tasks and recent notifications. 5. Teacher selects "Logout"; system ends the session and returns to the login page.

Alternative / exception flows	A1 (invalid credentials): System rejects the login and shows an error message. No session is created. A2 (session expired): System redirects the teacher to login before showing any protected page.
Postconditions	Success: an authenticated session exists and the dashboard is shown. After logout, the session is invalid and login is required again.
Business rules	Passwords are stored with a recognized secure hashing algorithm (NFR-03). All protected pages require authentication (NFR-03). Logout invalidates the session (NFR-04). Data is sent over HTTPS in deployed environments (NFR-05).
Relationships	Included by every other teacher use case (shown once in the use case diagram by a note).

UC-T02: Select course & section

Use case ID	UC-T02
Use case name	Select course & section
Source requirements	FR-03, FR-04; US-T2
Primary actor	Teacher
Secondary actors	System Administrator (maintains course, section and enrollment data)
Description	The teacher selects one of the courses and sections assigned to them and views the class schedule and enrolled students.
Preconditions	UC-T01 completed. The administrator has assigned courses and sections to the teacher.
Trigger	The teacher opens the Courses page.
Main success scenario	1. System lists the courses and sections assigned to the logged-in teacher. 2. Teacher selects a section. 3. System displays the class schedule and the enrolled student list for that section.

Alternative / exception flows	A1 (unassigned section): If the teacher requests a section that is not assigned (for example by changing a URL), the system denies the request and records the attempt. A2 (no assignments): System shows an empty-state message and no sections.
Postconditions	The selected section becomes the working context for UC-T03, UC-T04, UC-T05, UC-T07, UC-T08, UC-T09 and UC-T12.
Business rules	A teacher can access only the courses and sections assigned to them (FR-04, Constraints). Unauthorized access attempts are denied (NFR-04).
Relationships	Precedes UC-T03, UC-T05, UC-T07, UC-T08, UC-T09 and UC-T12.

UC-T03: Take attendance

Use case ID	UC-T03
Use case name	Take attendance
Source requirements	FR-05, FR-06, FR-07; US-T3
Primary actor	Teacher
Secondary actors	Students (subject of the record)
Description	The teacher marks each enrolled student as Present, Absent or Late for a class session and submits the attendance for the whole class at once.
Preconditions	UC-T02 completed. The class session date and section are known.
Trigger	The teacher opens attendance for a class session.
Main success scenario	1. System shows the enrolled student list for the session. 2. Teacher marks each student as Present, Absent or Late. 3. Teacher selects "Submit Attendance". 4. System validates that every student has a status. 5. System saves the date, section, session, student and status for each student. 6. System confirms that attendance was saved.

Alternative / exception flows	A1 (attendance already exists): System blocks a duplicate record for the same student and session and offers to open the existing record. A2 (incomplete marking): System highlights unmarked students and does not submit. A3 (save fails): System shows an error and saves nothing partially (NFR-06). A4 (correction after submission, permitted only): Teacher opens the submitted record, changes a status and saves. System records the previous status, new status, date and editor.
Postconditions	Attendance for the session is saved. Corrections, if any, are recorded in the change history.
Business rules	Only one attendance record per student per class session (FR-06, NFR-03). Corrections are allowed only when the institution permits them (Assumptions). Attendance changes are auditable (NFR-10).
Relationships	Extended by correction flow A4. Its output is read by UC-T04.

UC-T04: View & update attendance records

Use case ID	UC-T04
Use case name	View & update attendance records
Source requirements	FR-08, FR-09; US-T4
Primary actor	Teacher
Secondary actors	None
Description	The teacher reviews past attendance by student, section and date range, sees totals, and identifies students with repeated absences or late arrivals.
Preconditions	Attendance records exist for the assigned section (UC-T03).
Trigger	The teacher opens Attendance History.

Main success scenario	1. Teacher selects a filter: student, section and/or date range. 2. System lists matching attendance records. 3. System shows totals of Present, Absent and Late for the selected period. 4. System highlights students with repeated absences or late arrivals. 5. Teacher, if permitted, opens a record and updates it; system saves and logs the change.
Alternative / exception flows	A1 (no matching records): System shows an empty-state message. A2 (no edit permission): Edit controls are hidden and the record is read-only.
Postconditions	Records are displayed. Any permitted update is saved and recorded in the change history.
Business rules	Teachers can view and update attendance only for their assigned sections (FR-04). Updates require the required permission (FR-04 SRS version). Open point: the threshold that defines "repeated" absence or lateness is not stated in the SRS and should be confirmed.
Relationships	Reads the output of UC-T03.

UC-T05: Post announcement

Use case ID	UC-T05
Use case name	Post announcement
Source requirements	FR-10, FR-11; US-T5
Primary actor	Teacher
Secondary actors	Students in the selected section (recipients)
Description	The teacher writes an announcement for one assigned section, optionally attaches a file, and publishes it. Only students enrolled in that section can view it.
Preconditions	UC-T02 completed. The section is assigned to the teacher.
Trigger	The teacher selects "New Announcement".

Main success scenario	1. Teacher enters the announcement text and selects the target section. 2. Teacher optionally attaches a file. 3. Teacher selects "Publish". 4. System stores the announcement with its publication date and time. 5. System restricts visibility to students enrolled in the section. 6. System confirms publication and triggers UC-T06.
Alternative / exception flows	A1 (file upload fails): System shows an error and does not publish until the upload succeeds or the attachment is removed. A2 (edit or remove): Teacher edits or removes their own announcement when permitted; the change is saved.
Postconditions	The announcement is visible only to students of the selected section.
Business rules	Only the assigned section can be selected (FR-04). Teachers can edit or remove only their own announcements (FR-11).
Relationships	Includes UC-T06 (notification).

UC-T06: Notify students

Use case ID	UC-T06
Use case name	Notify students
Source requirements	FR-12, FR-13; US-T6
Primary actor	System (triggered by publication events and deadline schedule)
Secondary actors	Students (recipients); Teacher (views delivery status)
Description	When an announcement or assignment is published, the system notifies the enrolled students and sends deadline reminders when enabled. Delivery status is tracked where the service supports it.
Preconditions	An announcement or assignment has been published (UC-T05 or UC-T07). The notification service is configured.
Trigger	A publish event occurs, or a scheduled deadline reminder is due.

Main success scenario	1. System creates one notification for each enrolled student of the section. 2. System sends each notification through the configured channel. 3. System records the status: generated, sent or delivered, where the service provides those states. 4. Before a due date, the system sends reminders when reminders are enabled. 5. Teacher views the delivery status for the announcement or assignment.
Alternative / exception flows	A1 (delivery failure): The notification is marked failed. The announcement or assignment remains published (FR-13). A2 (service not configured): The system keeps the notification in the app and logs that external delivery is unavailable.
Postconditions	Notification records exist with a delivery status. The original content is unchanged.
Business rules	Notifications go only to enrolled students of the relevant section (FR-12). Notification content must not expose confidential information (NFR-11).
Relationships	Included by UC-T05 and UC-T07.

UC-T07: Create & publish assignment

Use case ID	UC-T07
Use case name	Create & publish assignment
Source requirements	FR-14, FR-15; US-T7
Primary actor	Teacher
Secondary actors	Students (view published assignments)
Description	The teacher creates an assignment with title, instructions, total marks, due date and optional attachments, then saves it as a draft or publishes it to students.
Preconditions	UC-T02 completed. The section is assigned to the teacher.

Trigger	The teacher selects "New Assignment".
Main success scenario	1. Teacher enters title, instructions, total marks and due date. 2. Teacher optionally attaches worksheets or learning materials. 3. Teacher selects "Save Draft" or "Publish". 4. System validates required fields and that total marks are greater than zero. 5. On publish, the assignment becomes visible to students in the section. 6. System records creation and publication details and triggers UC-T06.
Alternative / exception flows	A1 (validation fails): System highlights the invalid fields and does not save or publish. A2 (saved as draft): The assignment remains hidden from students until published.
Postconditions	The assignment is stored as Draft or Published. Only Published assignments are visible to students.
Business rules	Draft assignments are hidden from students (Constraints). Total marks must be greater than zero (FR-14 SRS version).
Relationships	Includes UC-T06 (notification). Its submissions are reviewed in UC-T08.

UC-T08: Review student submissions

Use case ID	UC-T08
Use case name	Review student submissions
Source requirements	FR-16, FR-17, FR-18; US-T8
Primary actor	Teacher
Secondary actors	Students (submit work)
Description	The teacher views submitted, late and missing work for an assignment in one place, opens or downloads submitted files, and allows late submission or resubmission when authorized.
Preconditions	A published assignment exists for an assigned section (UC-T07).

Trigger	The teacher opens the submissions list for an assignment.
Main success scenario	1. System lists submissions grouped as Submitted, Late or Missing. 2. Teacher opens or downloads a submitted file. 3. If authorized, teacher allows a late submission or resubmission. 4. System updates the submission status and records the change.
Alternative / exception flows	A1 (file unavailable): System shows a clear error message that the file cannot be opened. A2 (unauthorized course): System denies access to another course's submissions.
Postconditions	Submission statuses are current. Any allowed late submission or resubmission is recorded.
Business rules	Submission time is recorded automatically (FR-17). Late status is calculated from the due date and the configured deadline rules (NFR-13). Users can access only submissions in their own courses (NFR-04).
Relationships	Follows UC-T07. Feedback on submissions is given in UC-T12.

UC-T09: Record quiz & exam marks

Use case ID	UC-T09
Use case name	Record quiz & exam marks
Source requirements	FR-19, FR-20; US-T9
Primary actor	Teacher
Secondary actors	None
Description	The teacher enters marks for quizzes, midterms and final exams for students in an assigned section, reviews them and submits them. Marks are stored separately for each assessment.
Preconditions	UC-T02 completed. The assessment and its total marks are defined.

Trigger	The teacher selects an assessment and section under "Marks".
Main success scenario	1. Teacher selects the assessment type (quiz, midterm or final). 2. Teacher enters the mark for each student. 3. System checks each mark is within 0 and the assessment total. 4. Teacher reviews and corrects marks as needed. 5. Teacher selects "Submit Marks". 6. System confirms that the marks are saved for that assessment.
Alternative / exception flows	A1 (mark above total or negative): System rejects the entry and highlights it. A2 (missing marks): System lists students without marks before submission. A3 (save fails): System shows an error and saves nothing partially.
Postconditions	Marks are stored per student, per assessment and per section.
Business rules	Marks must not exceed the assessment total (FR-20). Only authorized teachers can enter marks for their sections (FR-19, NFR-04).
Relationships	Extended by UC-T10 when a submitted mark is corrected.

UC-T10: Edit submitted results

Use case ID	UC-T10
Use case name	Edit submitted results
Source requirements	FR-21, FR-22; US-T10
Primary actor	Teacher
Secondary actors	Academic Authority (approves corrections to finalized results); Student (notified of changes)
Description	The teacher corrects a submitted mark after finding a marking error. The system records the change and, for finalized results, waits for approval before applying it.

Preconditions	Marks have been submitted (UC-T09). The teacher has edit permission for the section.
Trigger	The teacher selects a submitted mark and chooses "Edit".
Main success scenario	1. Teacher enters the corrected mark and, if the policy requires, a reason. 2. System records the previous mark, new mark, date and time, and editor. 3. If the result is not finalized, the system applies the correction. 4. If the result is finalized, the system sends the correction to the Academic Authority for approval. 5. After approval, the system applies the correction. 6. If the published result changed, the affected student is notified.
Alternative / exception flows	A1 (no permission): System denies the edit. A2 (approval rejected): The original mark is kept and the teacher is informed.
Postconditions	The correction is applied (or pending approval) and the full history is preserved.
Business rules	A correction that needs approval is not applied until it is approved (FR-22). Result history is kept for authorized review (NFR-10). Students are notified when a published result changes (FR-22).
Relationships	Extends UC-T09. Uses the approval decision of the Academic Authority.

UC-T11: Submit final grades

Use case ID	UC-T11
Use case name	Submit final grades
Source requirements	FR-23, FR-24; US-T11
Primary actor	Teacher
Secondary actors	Academic Authority (approves and publishes grades); Student (views published grades)

Description	The system calculates final grades from the approved grading policy. The teacher reviews the results and submits them. The university then approves and publishes them.
Preconditions	All marks for the section are entered and valid (UC-T09).
Trigger	The teacher selects "Calculate & Submit Grades" for a section.
Main success scenario	1. System calculates final grades from the approved grading policy for the course. 2. Teacher reviews marks and calculated grades. 3. Teacher selects "Submit Grades". Status changes to Submitted. 4. Academic Authority approves the grades. Status changes to Approved. 5. Academic Authority publishes the grades. Status changes to Published. 6. Students can view their grades once published.
Alternative / exception flows	A1 (missing or invalid data): System lists the problems and blocks submission. A2 (self-approval attempt): If approval belongs to another authority, the system denies the teacher's approval.
Postconditions	Grade status is Submitted, Approved or Published according to the university process.
Business rules	Grade statuses are Draft, Submitted, Approved and Published (FR-24). Only authorized users approve and publish (FR-23 SRS version). Students see grades only when released (Constraints).
Relationships	Uses the marks from UC-T09 and may be followed by a correction in UC-T10.

UC-T12: Provide feedback to students

Use case ID	UC-T12
Use case name	Provide feedback to students
Source requirements	FR-25, FR-26; US-T12

Primary actor	Teacher
Secondary actors	Student (receives feedback)
Description	The teacher writes individual feedback for a student, linked to an assignment or assessment, and releases it when ready. Feedback is private to that student.
Preconditions	The student has a submission or result for the assignment or assessment.
Trigger	The teacher opens a student's submission or result and selects "Add Feedback".
Main success scenario	1. Teacher selects the student and the linked assignment or assessment. 2. Teacher writes feedback and saves it. 3. Teacher releases the feedback to the student. 4. Student opens the feedback from their account.
Alternative / exception flows	A1 (edit feedback): Teacher updates feedback when permitted; the update is saved. A2 (another student requests it): System denies access.
Postconditions	Feedback is stored and linked. It is visible only to the intended student after release.
Business rules	One student's private feedback is never visible to other students (FR-26, NFR-11). Only authorized teachers can create or update feedback for their sections (FR-25).
Relationships	Related to UC-T08 (submissions) and UC-T09 (assessments).

UC-T13: Report student issue

Use case ID	UC-T13
Use case name	Report student issue
Source requirements	FR-27, FR-28; US-T13
Primary actor	Teacher
Secondary actors	Academic Authority (receives and acts on reports)

Description	The teacher sends a confidential report about a serious academic, attendance or behavioural issue to the designated authority and can check its status.
Preconditions	The teacher knows the student's details. The designated authority is configured.
Trigger	The teacher selects "Report Issue" for a student.
Main success scenario	1. Teacher selects the student and an issue category (academic, attendance or behavioural). 2. Teacher enters a description and attaches supporting information. 3. Teacher submits the report. 4. System routes the report to the designated authority and records the date and submitter. 5. Teacher checks the report status where status tracking is supported.
Alternative / exception flows	A1 (required field missing): System highlights the missing field and does not submit. A2 (unauthorized access): Users who are not authorized cannot open the report.
Postconditions	Report status is Submitted. Only authorized users can access it.
Business rules	Reports are visible only to authorized users (FR-28, NFR-11). Confidential information is protected from unauthorized access (FR-27).
Relationships	Routed to the Academic Authority.

UC-T14: View schedule & pending tasks

Use case ID	UC-T14
Use case name	View schedule & pending tasks
Source requirements	FR-29, FR-30; US-T14
Primary actor	Teacher
Secondary actors	None

Description	The dashboard lists upcoming classes, deadlines, submissions to grade, marks awaiting entry and results awaiting action. The teacher can open any task directly.
Preconditions	UC-T01 completed.
Trigger	The teacher opens the dashboard or returns to it.
Main success scenario	1. System displays upcoming classes and relevant deadlines. 2. System lists pending submissions, unsubmitted marks and results awaiting action. 3. Teacher selects a task to open it directly. 4. After completing the task, the teacher returns to the dashboard. 5. System refreshes the task list to reflect the completed status.
Alternative / exception flows	A1 (no upcoming classes or pending tasks): System shows an empty-state message.
Postconditions	The task list reflects current status. The dashboard is read-only apart from navigation.
Business rules	The task list updates when a task is completed or its status changes (FR-30). Dashboard loads within 3 seconds under normal conditions (NFR-01).
Relationships	Entry point for UC-T02 through UC-T13.

4. Assumptions and open points

The institution decides whether teachers may correct attendance after submission, and under what conditions (SRS Assumptions).

The institution decides which authority approves finalized results and publishes grades (SRS Assumptions).

The threshold that defines repeated absence or late arrival (UC-T04) is not stated and should be confirmed.

The late-submission rules and any resubmission limits (UC-T08) come from institutional policy.

Email, SMS and other external delivery (UC-T06) depend on the notification service being configured.